Metaphysics seminar **10/3/03**

A modal analogue of the doctrine of temporal parts

the *x*s **compose** *y* iff each of the *x*s is part of *y*, and every part of *y* shares a part with one of the *x*s *x* and *y* **coincide** iff there are some things that compose *x* and compose *y*.

Some eternalists—orthodox mereologists like Sider and Lewis—think that the notion of parthood *simpliciter* makes sense, and that '*x* is part of *y*' entails '*y* exists whenever *x* does'. For them, the doctrine of temporal parts can be simply stated (top left box below.)

Other eternalists either deny that the notion of parthood simpliciter makes sense, or deny (like Fine and the A-theoretic eternalists considered by Sider in section 4.6) that it works the way the orthodox mereologists think it does. They can also state a version of the doctrine of temporal parts, but to do so they must use the temporally relativised predicate '*x* is part of *y* at *t*'. (middle left box)

Finally, presentists do think the notion of parthood simpliciter makes sense, but reject claims that entail the existence of entities not located at the present. They too can state a version of the doctrine of temporal parts. This presentistically acceptable claim can be derived in a fairly mechanical way from the eternalist version in the middle left box — so it would be natural for presentists to allow themselves this sort of talk, as a manner of speaking.

The debate between presentists and eternalists is in many ways analogous to the debate between *actualists* and *possibilists* in the philosophy of modality. (One point of disanalogy: almost everyone is an actualist.) The doctrine of temporal parts has a modal analogue; how we phrase it will depend on whether we're possibilists, and if so, whether we're happy with a notion of parthood simpliciter on which the claim that *x* is part *simpliciter* of *y* entails that y exists at every possible world where *x* does.

Doctrine of temporal parts	**"Doctrine of modal parts"**
Eternalist version, without relativisation: If *x* exists at *t*, then *x* has a part that exists only at *t*, and overlaps every other part of *x* that exists at *t*.	*Possibilist version, without relativisation:* If *x* exists at *w*, then *x* has a part that exists only at *w*, and overlaps every other part of *x* that exists at *w*.
Eternalist version, with relativisation: Whenever *x* exists at *t*, there is a *y* such that *y* coincides at *t* with *x*, and *y* exists only at *t*.	*Possibilist version, with relativisation:* Whenever *x* exists at *w*, there is a *y* such that *y* coincides [permanently] at *w* with *x*, and *y* exists only at *w*.
Presentist paraphrase: Always, for all *x*, there is a *y* such that *y* coincides with *x*, and *y* will not exist henceforth, and never existed hitherto.	*Actualist paraphrase:* Necessarily, for all *x*, there is a *y* such that *y* coincides [permanently] with *x*, and *y* has all its properties essentially.

The doctrine of temporal parts is often associated with another, stronger doctrine, which allows temporal parts to be combined in any old way into new objects. From the standpoint of an eternalist who believes in non-relativised parthood, this stronger doctrine follows from the doctrine of temporal parts together with the principle of **Unrestricted Composition** (Universalism): whenever there are some things, there is something they compose. But from the standpoint of the relativising eternalist, or the presentist, there is no such entailment.

Strengthened doctrine of temporal parts	**Strengthened doctrine of "modal parts"**
Eternalist, no relativisation: For any nontrivial partial function f from times to things, there is an x which is composed of all the y's such that y is the temporal part at t of f(t).	*Possibilist, no relativisation:* For any nontrivial partial function f from worlds to things, there is an x which is composed of all the y's such that y is the modal part at w of f(w).
Eternalist, with relativisation: For any nontrivial partial function f from times to things which exist at those times, there is an x which exists only at times in the domain of f, such that whenever f(t) = y, x coincides at t with y.	*Possibilist, with relativisation:* For any nontrivial partial function f from worlds to things which exist at those worlds, there is an x which exists only at worlds in the domain of f, such that whenever f(w) = y, x coincides at w with y.
Presentist paraphrase: Always, for any instantiated property p that is never instantiated by more than one thing, there is an x such that always, x exists only if something has p, and everything that has p coincides with x.	*Actualist paraphrase:* Necessarily, for any instantiated property p that is necessarily instantiated by at most one thing, there is an x such that necessarily, x exists only if something has p, and everything that has p coincides with x.

Yablo

Yablo doesn't actually explicitly endorse the strengthened doctrine of "modal parts", although he does at least endorse the doctrine that you get when you interpret 'coincident' in the DMP as 'spatially and temporally colocated'. He aims a view of the same sort, but generalised so as to make no special reference to spatiotemporal of mereological notions, and hence applicable to entities of many different ontological categories.

Definitions:
An **attribute** is a function from worlds to possible individuals. A **property** is an attribute such that anything that has it at all worlds where it exists, has it at all worlds whatsoever. [!? Fortunately this is pretty much irrelevant: for any predicate F, 'F or nonexistent' will express a Yablo-property.]

Where X is a set of properties, □ is an **X-refinement** of □ (‘□ ≥ □’) iff whenever □ has a property in X essentially, □ does too. [Note for cognoscenti: this doesn’t need to be relativised to worlds, since we’re assuming an S5 logic]

X is **upward-closed** iff at any world at which something exists that has certain properties in X, something exists that has those properties essentially.

X is **downward-closed** iff whenever □ is an X-refinement of □ that has certain properties in X essentially, necessarily, if □ exists, then □ exists and has those properties.

X is **closed** iff X is upward-closed and downward-closed.

A property P is **X-categorical** iff whenever □ is an X-refinement of □, necessarily, □ has P iff □ has P

Two things are **X-coincident** iff they have the same X-categorical properties.

X is **full** iff for any partial function f from worlds to objects which exist at those worlds, there is an x that exists only at worlds in the domain of f, such that whenever $f(w)=y$, x X-coincides at w with y.

X is **maximal closed** iff X is closed, and X is not a proper subset of any other closed set of properties.

X is **separable** iff no two things exist at all the same worlds, and are X-coincident in every world in which they exist.

Theorems:

Prop. 1: If X is closed, and □ is an X-refinement of □, then::

(i) at every world where □ exists, □ exists;

(ii) at every world where □ exists, □ has all the X-properties that are essential to □

(iii) at every world where □ has all the X-properties that are essential to □, □ exists.

Prop. 2: If X is closed, every member of X is X-categorical.

Prop. 3: If X is maximal closed and full, then every X-categorical property is a member of X. [Hence X is closed under negation, conjunction, etc.]

Prop. 4: If X is closed and separable, then no two things have all the same X-properties essentially.

Terminological note: Yablo does the whole thing relative to a model. And instead of saying things like ‘X is upward-closed in *W*’, he says ‘<*W*,X> is upward closed’.

So what claims is Yablo making?

- There is a set of properties that is maximal closed, full, and separable — call them the categorical properties.
- The categorical properties are the properties that are ‘grounded entirely in how it is in the circumstances that happen to obtain’, by contrast with the hypothetical properties, which are ‘grounded not just in how a thing *actually* is, but on how it *would* or *could* have been if circumstances had been different’.

- The categorical propeties are the *cumulative* properties: the ones which 'can only "build up" the essences in which they figure', as opposed to *restrictive* properties—e.g. *being identical to x*, or *being a cloth*, such that by including one of them in an essence we 'block the entry of certain of its colleagues'. [Worry: *being identical to x* doesn't seem *hypothetical*]
- Many ordinary predicates—not just 'size, weight, color and so on' but also predicates like 'having enshrouded Jesus Christ'—are categorical
- ??? There is only one maximal closed, full, separable set of properties ???
- There is an important relation of "coincidence" — which Yablo provocatively and misleadingly likes to call "contingent identity" — defined as sharing all the same categorical properties.
- The fact that, e.g., the statue "coincides" in this sense with the clay *explains* why they are so similar in so many respects, a fact that would otherwise be inexplicable.